

BOOK IX — TESTS AND PREDICTIONS

"WORKING DRAFT — part of the GRUT working corpus; statuses per `books/CORPUS_CHARTER.md`; subject to chapter-by-chapter audit; nothing here banks."

GRUT Books — Working Edition v1.0 · D. Ryan Grover · 2026-09-07 · release `zenodo-v1.0` · Markdown is the authoritative source; the PDF is generated from it. Drafted from the frozen record by the RAI builder (Claude, Anthropic) under owner direction — this edition is a publishing pass only: formatting, navigation, and metadata; no claim strengthened or weakened for publication.

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1 · The organizing fact: the PREDICTED section is empty

Every book in this corpus reserves the word *prediction* for gate-earned results, and this is the book that governs entry. Its organizing fact can therefore be stated in one line, at the top, where it belongs:

The PREDICTED section of the GRUT record is empty. Nothing has earned entry.

STATUS: EMPTY (nothing has earned entry; Book IX governs entry) — canonical claim 21, verbatim (source: books/CORPUS_CHARTER.md ; GRUT_PROGRAM_FREEZE.md §3, "PREDICTED — EMPTY, with the reason on the record").

This is not an apology and not a placeholder. It is a *result*, and it comes with a derived reason, established at two levels of the record. At the structural level: the framework's commitment is a **class** of admissible response kernels, a class has no scale, and the admissible set is an amplitude-homogeneous cone — the passivity/cone theorems orient channels and pin no ratio, so "every route from this framework to a number runs outside it" (GRUT_PROGRAM_FREEZE.md §3). At the level of the one candidate carried all the way through a gate: the Γ_T design table (§5 below) exposed the same fact in ledger form —

Everything that is fixed lives where nothing is observable; everything observable-sized runs through unresolved inputs. In the gate's own words: "every quantity that is fixed lives in horn (a); every quantity that could make Γ_T observable lives in horn (b) and is unresolved. The two horns do not mix: no fixed coefficient feeds the observable-sized number, and no observable-sized number is fixed."

STATUS: DERIVED (status-level: read off the gate's fixed-versus-free table and the freeze's cone theorems; no physics evaluated in the reading) — the derived reason the PREDICTED set is empty (source: GRUT_PREDICTION_GATE_GAMMA_T.md §5; GRUT_PROGRAM_FREEZE.md §3).

Nor is the emptiness for lack of searching. Two FOREST discriminator hunts returned FOREST-EMPTY ("no live register node is a working discriminator" — the second hunt confirming the first by an independent route); the signature audit returned EMPTY across every candidate observable; and the X_FLOOR campaign, built to ask whether anything *pins* the interior modulus x , landed NO PIN on its computed route (R1: the anomaly's own contribution ≥ 108 decades below every banked bound, the scheme piece free — "that freedom is the no-pin finding") and closed its dynamics route as a classifier (the channel-diagonal passivity lemma: a convex cone, no ceiling, no pin) (source: GRUT_PROGRAM_FREEZE.md §3; PHYSICS_LEDGER/FOREST_PHASE11_MAPPING.md ; X_FLOOR_MAP.md ; calc/RESULTS_anomaly_c0_map.md ; calc/RESULTS_x_no_pin.md).

The rest of this book is the record of how that emptiness was established, policed, and priced: the philosophy that assigns the votes (§2), the entry rules (§3), the gate methodology as demonstrated (§4), the Γ_T closure in full (§5), the signature audit (§6), the candidates still standing without gates (§7), the graveyard of failed routes (§8), and the future gates and reopening keys (§9).

2 • The experimental philosophy

What counts as a prediction. On this record, a prediction is a gate-earned result: a quantitative consequence that is nontrivial and not already encoded in the inputs, reached through a pre-registered design that survives all five discrimination requirements (§4) and whose computed output then survives the program's own screens. Nothing else may carry the word. In particular, the mandatory honesty note of the freeze applies throughout: the framework reproduces standard results *because its executed machinery is standard machinery on the declared inputs* — recovery-by-identity is compatibility, not

correspondence evidence, and **nothing RECOVERED counts toward PREDICTED** (source: GRUT_PROGRAM_FREEZE.md §3, RECOVERED note; GRUT_NEXT_STEPS.md, the prediction hunt).

Signatures versus no-go directions. The record's standing distinction, kept on the face of both ledgers: a *signature* is a positive distinctive observable; a *falsifiable-direction no-go* is a statement about what the framework forbids. GRUT currently has the second and not the first. The no-go ledger is explicit about strength calibration — its legend (FORBIDDEN / SETTLED-NEGATIVE / EMPIRICALLY EXCLUDED / INVISIBLE-BY-SUPPRESSION / BORROWED) is load-bearing, and the register currently banks **no genuine FORBIDDEN entry at all**; reporting that absence, rather than promoting a settled-negative to an impossibility, is itself part of the discipline (source: SIGNATURE_AUDIT.md header; NO_GO_LEDGER.md legend and closing note).

The standing kill condition. The framework's cleanest exposure to data is a direction it shares with an entire class: no purely relaxational kernel crosses $w = -1$; only an oscillatory pole pair does. Any observed crossing excludes the whole passive relaxational family at a stroke, GRUT included — the freeze elevates this to a formal reopening condition ("Nature votes at $w = -1$ ").

STATUS: DERIVED (class-level; explicitly not GRUT-specific) — canonical claim 15, verbatim: menu-scope exclusion shared by the entire passive class (source: books/CORPUS_CHARTER.md ; GRUT_MODEL_FRAMEWORK.md §5; GRUT_PROGRAM_FREEZE.md §5, condition 4).

The posture. The model framework's §8 states both lists — what would strengthen GRUT (the zero-mode returning *protected*; a §2-selection derived from an independent principle; a gate-surviving sector consequence; the SLOT test discharging the clock-slot) and what would weaken or falsify it (an observed $w = -1$ crossing; the zero-mode returning *lifted*; a self-closing description of the time-asymmetry; the USL shape signature excluded in its distinguishable regime; the RESIDUE test deriving the alignment from unoriented hypotheses) — and then closes with the sentence this book adopts as its own summary of the experimental philosophy: **"Nature and mathematics hold every one of these votes. The framework holds none of them."** (source: GRUT_MODEL_FRAMEWORK.md §8, verbatim).

The stop rule. Behind the posture sits committed governance: no new generation, model class, or foundational rung may be created merely because the previous one failed; and if no discriminator survives the comparative pass and the prediction hunt, the program says so in the model document and stops — no automatic rescue, no new rung. The reopening keys are hung where the program cannot reach them by itself (source: GRUT_PROGRAM_FREEZE.md §1; GRUT_NEXT_STEPS.md, stop rule).

3 · The entry rules: the preregistration discipline

The methodology below is enforced, not aspirational, and it exists because the program caught itself violating it — repeatedly, in one recurring shape.

Immutability after hashing. Every pre-registration is immutable once hashed (provenance/prereg/MANIFEST.txt carries the seals); results live in **separate files** citing the seal in the pinned `sha256 = <64hex>` format; editing any sealed file breaks the verification check. And the record is explicit about what a green run means: the *discipline* holds — declarations complete, seals intact,

guards firing. "It does not certify physical truth — a wrong-but-well-provenanced claim passes, and the register says so about itself" (source: `HOW_TO_VERIFY.md` §§3–4).

The self-certification pattern. The reason results must live outside the files that pre-register them is that the program's audits found multiple defects with a single structure: *the thing that certifies sitting inside the thing being certified* — a selftest whose answer lived in its own print statement; a ratchet computed from the very list it existed to bound; results appended to the pre-registration they were meant to be tested against. Each was caught only by an outsider constructing a case the author's own controls had already certified as clean. The remedy adopted program-wide: move the certifying thing outside — an external hash, a hardcoded literal, a separate file, an adversary who did not write the answer (source: `STAGE_CLOSE_2026-08-09.md`; `AGENT_COORDINATION.md`, the "standing self-certification pattern").

The instrument standard. The freeze's §7 lists what the method earned as its transferable asset: pre-registration with hashes before results; **two-sided gates**; negative controls to the N1–N10 standard; **journal-read, de-pinned instruments** (an instrument must extract the statuses it adjudicates from the record at runtime, never carry them pinned in its own text); mirror controls; preserve-the-failure; the stopping rule itself. The same section records the price of learning this: **the audit apparatus itself was caught three times** — pass-label gates, story-pinned gates, the uncalibrated sorter — each generation of instrument caught, stamped, and preserved rather than erased (source: `GRUT_PROGRAM_FREEZE.md` §§4, 7).

The pass-label lesson, recurring and caught again. The pass-label pattern — a gate whose "pass" is defined so that only the desired answer can pass — recurred in the Γ_T closure instrument itself and was caught pre-commit by the SPEC-compliance checker: the script contained a margin *gate* that hard-failed unless the result landed ≥ 50 orders below the bound. It was rebuilt as a report-never-gate branch, and the final instrument's gates "assert well-formedness, arithmetic identities, and extraction success only — no gate asserts which verdict passes, and magnitude findings are reported, never gated" (source: `calc/RESULTS_gw_tensor_friction.md`, adversarial verification, defect 1 and closing line).

The discipline demonstrated on a data gate. The low- ℓ TT-auto gate on the interior modulus x is the record's fullest demonstration that the firewall machinery works on data-facing instruments too: the first frozen version was **voided the same day by the program's own firewall** (a Bessel-recurrence bug made every headline constant noise — the "edge" moved by a factor of ~ 8 on one resolution doubling), then rebuilt, verified against an independent harness, and legally re-frozen under a written amendment record A1–A7.

STATUS: DERIVED (instrument-grade data gate, re-frozen after its first freeze was voided by its own firewall; **unconditional most-conservative-corner bound** $x < 0.358$; **the κ -filter is the entire systematic**; **the A6 Boltzmann-grade differential check is OWED before any $\kappa=3$ kill-grade use**) — (source: `calc/RESULTS_isw_tt_auto.md`; `calc/isw_tt_auto.py`).

The entry rule this section adds up to. For a claim to enter the PREDICTED set it must pass through a design-first gate of the §4 kind, reach DISCRIMINATING on all five requirements, and then have its computation survive the same screens as everything else — the SPEC template says it on its face: "Nothing here banks; the output needs the four-lens screen and the bank gate like anything else" (source: `calc/SPEC_gw_tensor_friction.md` preamble). No such claim currently exists.

4 · The prediction-gate methodology, as demonstrated

The record contains exactly one prediction gate executed end-to-end: `GRUT_PREDICTION_GATE_GAMMA_T.md` (2026-09-06, owner-ordered), run **design-first** — every verdict-shaped statement preceding computation is derived from the *statuses* of register inputs, not from values. Its structure is the program's demonstrated protocol for any future candidate.

The eleven steps, as the gate instantiates them:

1–2. Read the record — every source the candidate touches, listed on the gate's face. 3. Name what the candidate connects — which registers of the program the slot joins. 4. Write the governing equation *before any calculation*. 5. Write the baseline — GR, and every standard parameterization the slot is degenerate with. 6. List **every** input entering the calculation, each with its register status. 7. Sort fixed versus free — what is genuinely pinned, what is staked/inserted/chosen. 8. Establish the licensed frequency/IR domain — where the record permits evaluation at all. 9. Test absorbability — can existing parameters (α_M , the free open-EFT slot) mimic it? 10. Pre-register the decision tree, before any evaluation. 11. Render the gate verdict — at design time, from statuses alone, where that is decidable.

The five-outcome decision tree (step 10, declared before evaluation): verdicts are **DISCRIMINATING** / **DEGENERATE** / **UNDERDETERMINED** / **OUT OF VALID DOMAIN** / **NO EFFECT**, rendered **per horn and never composed into a single label**, with **UNDERDETERMINED** and **OUT OF VALID DOMAIN** taking precedence over magnitude verdicts for anything they touch. The rules: **R1 (sector)** — the friction-carrying structure must be *derived* into the channel from booked nodes; entry by projector choice caps the verdict at **UNDERDETERMINED**. **R2 (domain)** — any load-bearing kernel feature supported at $\omega \leq 3.4H$ is **OUT OF VALID DOMAIN**. **R3 (fixedness)** — any staked amplitude or unverified identification in the chain forces **UNDERDETERMINED**. **R4 (absorbability)** — distinguishable from α_M only is class-level, not **DISCRIMINATING**; indistinguishable from the free SCDP slot is **DEGENERATE**. **R5 (magnitude)** — more than 10 orders below both the slot bound and any named future sensitivity is **NO EFFECT**. **DISCRIMINATING requires all five at once**. Beneath the gate, the calc-level SPEC carries its own four pre-registered outcomes — **PASS** (parameter-free bridge) / **FAIL-BUT-INFORMATIVE** (relocation, priced +1 at entry) / **CLOSES-THE-QUESTION** (channel empty) / **REFUSE** (the sector question undecidable from the booked family) — and the gate maps them into the tree explicitly (source: `GRUT_PREDICTION_GATE_GAMMA_T.md` §8; `calc/SPEC_gw_tensor_friction.md` §5).

The wording rule (owner correction, adopted on the gate's face): a candidate under a gate is **the currently identified candidate** for a nontrivial cross-sector consequence — never "the only place such a thing could live." Nothing forces the discriminator to live in the slot being gated; the gate adjudicates that candidate only (source: `GRUT_PREDICTION_GATE_GAMMA_T.md` preamble).

STATUS: DERIVED (methodological record: the protocol exists as executed governance, not as a claim about nature) — the eleven-step design-first gate with a five-outcome preregistered decision tree and per-horn routing (source: `GRUT_PREDICTION_GATE_GAMMA_T.md`, whole document).

5 · The Γ_T closure, in full

This is the record's one complete pass from candidate to computed closure, and every part of it is load-bearing for how this book's organizing fact was earned.

5.1 The candidate

Γ_T — a dissipative friction term in the cosmological tensor wave equation — was the currently identified candidate for a nontrivial cross-sector consequence, because one slot connects three registers of the program: (1) the **derived Tier-4 TT dissipation kernel** (gravity sector); (2) the **KMS/FDT lock** (any friction must arrive with its stochastic companion at the $\coth(\omega/2T_{dS})$ ratio, $T_{dS} = H/2\pi$); (3) **cosmological tensor propagation**, where the SCDP open EFT (arXiv:2507.03103) has parameterized the same object class and the only measured statement exists: $|\Gamma_T| \lesssim \text{few} \times H_0$ on the shared slot. The pre-registered question, unchanged from the register: *does the local memory scale connect to the cosmological friction parameter-free, or does the bridge need a new inserted scale?* (source: GRUT_PREDICTION_GATE_GAMMA_T.md §1; calc/SPEC_gw_tensor_friction.md §2).

The record contains two kernel candidates — the gate's two horns, which never mix:

- **Horn (a), the DERIVED kernel** (Tier-4, flat contract scope, $\omega \gg H$): chromatic, $\Gamma_T \propto \omega^3$, parameter-free, sign fixed by passivity.
- **Horn (b), the BOOKED two-scale ansatz** (calc/wz_dark_energy.py, register rung7_wz): $\chi(\omega) = A/(1 - i\omega\tau_c) + B/(1 - i\omega\tau_2)$ with $\tau_2 \sim 1/H_0$ — achromatic in the observing band, $\Gamma_T \rightarrow B \cdot H_0/2$.

5.2 The design-time verdict

The gate's step-6 input table graded all twelve inputs; step 7 sorted them, exposing the fixed/observable disjunction quoted in §1. Step 8 licensed the domain: the ground-based band sits at $\omega/H_0 \sim 10^{20}$, fully licensed for horn (a) (the derived refusal boundary $\omega = 3.3993H$ is 20 orders below the band), while horn (b)'s τ_2 pole is an analytic feature at $|\omega| \sim H_0$ — inside the region ROOT-1 established as **UNASKABLE** on four independent obstructions (no frequency variable is even defined there). Step 11 then routed what was decidable from statuses alone:

- **Horn (b) trips R1, R2 and R3 simultaneously** — the sector assignment is CHOSEN, the pole sits in the unaskable region, B is staked and $B \equiv \epsilon$ unverified — and even resolved, its achromatic form is R4-DEGENERATE with the free SCDP slot. *No route from horn (b) to DISCRIMINATING exists on the current record.*
- **Horn (a) passes R1–R3** (derived, licensed, parameter-free) **but is foreseen dead at R5**: its scale against the slot bound is set by $(\omega/\omega_{Pl})^2$, placing it ≈ 60 orders below $\text{few} \times H_0$ at design-level foresight. Because horn (a) has no free parameter, a calculation can only confirm this routing, never move it. **"A calculation whose verdict cannot change is a closure, not a hunt."**
- The one in-principle discriminating structure — the chromatic ω^3 term with fixed coefficient plus the KMS-locked noise companion — lives entirely in horn (a) and dies with it at R5; its relation-level content (dissipative $\neq \alpha_M$; noise mandatory) is class-level and already fenced against GRUT credit.

STATUS: CLOSED (gate outcome, rendered at design time from register statuses alone: the design does not survive as a prediction hunt; no number computed in the verdict) — (source: GRUT_PREDICTION_GATE_GAMMA_T.md §9).

5.3 The closure computation

The gate left one honest move available — a closure computation, explicitly priced as "a closure, not a prediction" — and the owner authorized it. `calc/gw_tensor_friction.py` was built to the **unmodified 2026-08-22 SPEC** (the immutable calc-level pre-registration), to the de-pinned standard: its verdicts are computed at runtime from statuses extracted out of `provenance/claims.json` and `PHYSICS_LEDGER/ROOT1_KERNEL_ORIGIN.md`, never carried in the script's own text.

SPEC outcome: REFUSE, with the obstruction stack named. Q-A — the sector question the SPEC says dominates everything else — cannot be settled from the booked family: every booked level books the channel assignment as a projector **choice** (p_tt CHOSEN, five-angle interrogation unanimous; the operator-basis frame CHOSEN at its enumerated order); the SPEC's two horns disagree; nothing booked selects between them. The full stack: (i) p_tt CHOSEN; (ii) frame-level CHOSEN; (iii) τ_2 INSERTED un-sourced (+2 of rung7's +3); (iv) the pole's region $\omega \leq 3.4H$ UNASKABLE — the derivation that would settle Q-A cannot currently be posed (source: `calc/RESULTS_gw_tensor_friction.md`).

The pre-registered question, answered: NO parameter-free connection exists on the licensed record. The only parameter-free, licensed entry into the slot is the Tier-4 derived kernel's

$$\Gamma_T(\omega) = (3/1280\pi) \cdot (\omega^3/\bar{M}_P^2) \cdot [1 + (104/9)(H_0/\omega)^2] \quad (\text{chromatic} \propto \omega^3; \mu\text{-independent})$$

giving $\Gamma_T/H_0 = 6.19 \times 10^{-63}$ at 100 Hz — **62.7 orders below the shared-slot bound** $\text{few} \times H_0$ (arXiv:2507.03103), final at these declarations because the horn has no free parameter. The gate's NO EFFECT routing is thereby **computed, not foreseen**.

STATUS: CLOSED (computed NO EFFECT; SPEC outcome REFUSE on the observable route; commits 2116251, 41e1af5) — canonical claim 16, verbatim: the Γ_T parameter-free value ($6.19e-63 \cdot H_0$ at 100 Hz) (source: `books/CORPUS_CHARTER.md`; `calc/RESULTS_gw_tensor_friction.md`; `calc/gw_tensor_friction.py`; `GRUT_PREDICTION_GATE_GAMMA_T.md`).

The conditional exhibits, labelled and unbanked. The SPEC's owed composition of horn (b)'s two live B values was performed and never composed silently: $B = 0.4$ (staked) gives $\Gamma_T = 0.20 \cdot H_0$ and a standard-siren amplitude effect $\Xi(z=0.5) - 1 \approx 3.7 \times 10^{-2}$; $B \approx 2.4 \times 10^{-4}$ (the conformalon leg) gives $1.2 \times 10^{-4} \cdot H_0$ and 2.2×10^{-5} — 3.2 orders apart, with nothing on the record selecting between them. The match-temptation fence was applied in full: the staked $B = 0.4$ landing inside the slot bound carries **evidential weight zero**. The $B \equiv \varepsilon$ identification is carried UNVERIFIED; no computation uses it. The unpinned ω_c moves the UV/IR crossover across a 19.8-order span (9.8×10^{-9} Hz / 22 Hz / 0.64 THz on the three in-corpus values) and enters no headline (source: `calc/RESULTS_gw_tensor_friction.md`, conditional exhibits; `calc/SPEC_gw_tensor_friction.md` §§3–4).

The hostile verification. Three independent checkers ran pre-commit. Arithmetic: a from-scratch recomputation reproduced every table value to the printed digits. SPEC compliance: four minor defects

caught and fixed, disclosed per house rule — including the pass-label margin gate of §3. And a **hostile refutation of the REFUSE**: an exhaustive search of all 74 register nodes found **no FORCED verdict anywhere** — three nearest misses, each failing on the register's own face (noise-transversality does not force tracelessness, "the TT restriction remains CHOSEN"; `response_lorentz_covariance` is TT-scoped at the vertex and corroborates the domain obstruction; the `rung9b` a/c sector-split assigns anomaly counterterms at k^4 , not the τ_2 pole). Obstructions (i)–(iii) satisfy the REFUSE clause even with (iv) struck (source: `calc/RESULTS_gw_tensor_friction.md`, adversarial verification).

Standing consequences. The signature audit's gate-to-readmit is closed **as a computed refusal** — the $\leq 10^{-21}$ figures stay un-readmitted, no number banks. EDIT 1's conditional marker is NOT finalised (that required a scalar-only Q-A answer; a REFUSE is not one). And under the freeze's stop rule, with the Γ_T candidate closed by computation, the honest program statement stands:

STATUS: CLOSED (gate outcome + closure computation: "no discriminator identified on the current record"; remaining candidates each require their own gate before any computation) — (source: `GRUT_PREDICTION_GATE_GAMMA_T.md` §9; `calc/RESULTS_gw_tensor_friction.md`, standing consequences).

6 • The signature audit record

`SIGNATURE_AUDIT.md` is the standing record of GRUT's empirical reach — every observable the program could in principle touch, and what an exhaustive audit found. Its headline is printed on its own face: at interpretation-level, GRUT is signature-null. The audited candidates, compressed:

candidate	audit verdict
GW dissipation (dephasing, $v_g \neq c$)	invisible-by-suppression (~ 22 – 62 orders below sensitivity)
dark-energy $w(z)$	signature-null (sourced = Λ CDM; evolving w needs an inserted scale)
tabletop decoherence (energy-basis wedge)	invisible-by-suppression (quiet-or-faint)
linear cosmology (μ , Σ growth)	signature-removing (no-go export: $\mu = 1$; GRUT forbids its own naive $\mu = 4/3$)
founding ζ / info / L_0 screens	no observable
linear-cosmology interior ({shear, bulk} family)	constrained-to-a-computed-window (the fourth category)

STATUS: CLOSED (audit outcome: EMPTY — no admissible, dedicated, parameter-free signature survives; scoped to GRUT-as-written, and known conditional on the CHOSEN projector per the audit's own post-interrogation note) — (source: `SIGNATURE_AUDIT.md`, audit-critic verdict and 2026-08-02 post-interrogation note).

Four items from the audit's verified 2026-08-02 extension are individually load-bearing for this book.

The scoped null. The external four-domain hunt (pre-registered A/B/C bar, adversarially refereed, numbers overseer-verified) found no candidate above Grade C in the **explored structure** — the admitted TT-channel / passive-KMS / Λ CDM-at-linear family. This confirms the EMPTY verdict from outside, and is explicitly *not* the universal claim "GRUT is empirically silent," which no search can establish.

STATUS: CLOSED (audit outcome: **scoped null — no candidate above Grade C in the explored structure; not a universal silence claim**) — (source: SIGNATURE_AUDIT.md , 2026-08-02 item 1).

The DESI anti-signature — the live threat. GRUT's sourced prediction is $w = -1$ flat; the passivity no-go forbids a single passive relaxor from crossing $w = -1$; DESI's preferred w_{w_a} trajectory crosses the phantom divide at $z \approx 0.35\text{--}0.5$ — exactly the forbidden shape. The four attribution fences travel with it, verbatim: (a) the preference is DESI BAO + CMB + SNe — DESI BAO alone shows no significant preference, and with a fixed r_d anchor the $\sim 3\sigma$ does not reproduce; (b) the honest headline is 3.1σ (DESI DR2+CMB, arXiv:2503.14738) — the 4.2σ endpoint rides the contested DESY5 compilation; (c) Bayesian model comparison gives only weak-to-moderate evidence; (d) the direction is Quintom-B, phantom in the past, quintessence today.

STATUS: pending refutation if the signal consolidates — the audit's verbatim standing disposition, with its comparative clause carried: "GRUT is structurally worse-placed to survive it than Λ CDM+quintessence" (source: SIGNATURE_AUDIT.md , 2026-08-02 item 2).

The geometry deserves one plain sentence, and Book VI's is adopted unchanged: there is no branch in which this observable delivers a confirming signature — if the crossing consolidates, the whole passive class is excluded; if it dissolves, GRUT's sourced cosmology remains indistinguishable from Λ CDM. The audit's word for this shape is *anti-signature*.

The α_M category fence and the Ξ_0 rule. Standard-siren friction α_M (running Planck mass) comes from a Hermitian action: removable by field redefinition, graviton-number-conserving, sign-indefinite, achromatic, noiseless. A genuine dissipative $\text{Im } \Sigma_R$ is none of those; they are slot-degenerate in the mean-field equation only. The consequences cut both ways: GRUT's object class is not absorbable into α_M — but a **detected $\Xi_0 \neq 1$ could never confirm GRUT**, and the distinction identifies the dissipative class, not GRUT within it.

STATUS: DERIVED (audit-level category distinction within standard theory; **no GRUT credit — the distinction identifies the dissipative class, not GRUT within it**) — $\alpha_M \neq \text{Im } \Sigma_R$ (source: SIGNATURE_AUDIT.md item 4).

The SCDP slot bound — the first measured bound on the object class. The Salcedo–Colas–Dufner–Pajer open EFT (JHEP 02(2026)241, arXiv:2507.03103) parameterizes a genuine dissipative tensor friction with mandatory stochastic source, explicitly distinct from $\alpha_M \cdot H$ — the first mainstream parameterization of the object class — and current friction measurements transfer as $|\Gamma_T| \lesssim \text{few} \times H_0$: **a no-cancellation bound on the shared slot, not a decomposed measurement of GRUT's kernel**, and no validation of any GRUT value (the retarded+noise form is u1-generic).

STATUS: ASSUMPTION (EMPIRICAL-INPUT, external, shared-slot; the retarded+noise form itself is u1-generic — no validation credit for the form) — (source: SIGNATURE_AUDIT.md item 5; arXiv:2507.03103 as the record cites it).

The one soft spot. Exactly one observable was never closed by a dedicated calculation: black-hole quasinormal modes / ringdown damping, where a lossy tidal response could in principle differ from GR's conservative one. The expectation on record — inheritance of rung4's structural Planck suppression — is an argument, not a computation, and the EMPTY verdict carries this one explicit caveat.

STATUS: UNRESOLVED (invisible-by-inheritance, not a dedicated calculation — the signature audit's single explicit caveat) — QNM/ringdown (source: SIGNATURE_AUDIT.md , "The one soft spot").

The NOT-banked fence, honored throughout this book: no $\leq 10^{-21}$ figure is a banked number — including reworded as "GRUT's effect" (no calculation exists; the figure in the graviton-mass demotion is arithmetic about a *test's reach*); and no "ET closes ~ 1.5 orders" claim (one choice inside a 2.5-order forecast spread). The fence's gate-to-readmit was `calc/gw_tensor_friction.py`; that gate is now closed **as a computed refusal**, so the fence stands permanently on the current record (source: SIGNATURE_AUDIT.md , NOT-banked; `calc/RESULTS_gw_tensor_friction.md` , standing consequences).

7 • Candidates still standing — none gated

Two candidates named by the gate's own closing paragraph remain on the record. Neither has a designed gate. The record is explicit that each "would need its own gate" before any computation, and this book states plainly that **no such gate design exists for either** — an absence, mapped, not a task silently assumed done.

7.1 The USL shape candidate

GRUT's noise kernel driving the Anastopoulos–Hu (2013) gravitational-decoherence master equation yields a signature distinguished from Diósi–Penrose/CSL **in shape only**: an energy-basis rate $\Gamma(\Delta E) = |A_{nm}|^2 S(\Delta E/\hbar)/\hbar^2$, scaling with the energy gap and ignoring spatial size — the qualitative opposite of position-basis collapse models — with a parameter-free predicted *shape* (suppressed rise, peak at $\Delta E = 1.22 \hbar\omega_c$, cutoff).

STATUS: HYPOTHESIS (proposed phenomenological relation; shape-only; magnitude verdict quiet-or-faint) — canonical claim 12, verbatim (source: books/CORPUS_CHARTER.md ; register node `rung8_falsifier`; GRUT_MODEL_FRAMEWORK.md §5).

The magnitude verdict is itself computed, and it is the reason this candidate does not carry the program: the dominant diagonal coupling ($T^{00} \sim$ energy density) commutes with H_S and samples $S(0) = 0$ — a quiet bath, $\Gamma = 0$; the wedge-carrying off-diagonal coupling ($T^0{}_i/T^{ij} \sim v/c$) survives but lands 7–47 orders below detectability; observability would require staking the noise amplitude $\sim 10^7\times$ above natural, a tuned number at the current matter-wave bound.

STATUS: CLOSED (computed outcome B — quiet-or-faint: diagonal coupling samples $S(0) = 0$, $\Gamma = 0$; the off-diagonal wedge is 7–47 orders below detectable; the falsifier does not carry the program) — (source: register node `rung8_falsifier` tier_note and differentiator; `calc/q1_energy_basis_magnitude.py`; `calc/energy_basis_decoherence.py`; `NO_G0_LEDGER.md` entry 4).

Two live threads could move it, both on the record and both open: the specialist gray-zone check (any leading-order off-diagonal energy coupling sampling $S(\Delta E)$ at $O(1?)$), and the +2 staked inputs made explicit (the amplitude normalization κ ; the cutoff ω_c — what "689 Hz" really was). The bar it must eventually meet is also on the record: the parameter-free Diósi–Penrose model was excluded at Gran Sasso — it made a magnitude claim and died by it. GRUT's shape claim, in its quiet-or-faint state, cannot yet be killed that way — which by the program's own lights is a deficit, not a safety (source: `books/BOOK_III_QUANTUM_REALITY.md` §7, consistent with `GRUT_MODEL_FRAMEWORK.md` §5). **Its eleven-step gate has not been designed; the record is silent on what its decision tree would be.**

7.2 The rung8 bookkeeping candidate

`GRUT_NEXT_STEPS.md` lists, after Γ_T and the USL shape, "the rung8 decoherence bookkeeping against tabletop bounds" as the third currently identified candidate. The record contains its inputs (the rung8 node's staked κ and ω_c , the declared cutoff convention, the matter-wave bound as the binding edge) but no gate design, no SPEC, and no pre-registered decision tree.

STATUS: UNRESOLVED (named candidate; no gate designed; nothing beyond the register node's own bookkeeping exists) — (source: `GRUT_NEXT_STEPS.md`, prediction hunt; register node `rung8_falsifier`, ledger_note and cutoff_convention).

7.3 The interior window — a bound, not a prediction

The one place where data currently binds a GRUT parameter is the linear-cosmology interior: the {shear, bulk} family's scalar admixture is bounded by measurement (DESI Σ_0 lensing: $x < \sim 0.59$ central-inputs, loose-upper per the F-MAP fence, hence $\mu - 1 \leq 0.20$), the audit's *fourth category* — constrained-to-a-computed-window rather than deleted by fiat. The null is not softened: **x has no floor, so no detection confirms GRUT and no null refutes it** — the family allows up to the edge and predicts nothing. The TT-auto gate (§3) is the standing instrument on this window, awaiting a computed candidate x^* that the record does not contain.

STATUS: UNRESOLVED (register tier to-derive, default-BROKEN; constrained-to-a-computed-window; x has no floor — no detection confirms, no null refutes) — harmonized with Book VI §VI.2.3 at corpus audit (source: `SIGNATURE_AUDIT.md` table row 6; `calc/RESULTS_isw_tt_auto.md`; `X_FLOOR_MAP.md`).

8 · Failed and refused prediction routes — the graveyard, preserved

The program's rule is preserve-the-failure, and this section is its exhibit hall. Every entry is part of model history, stated on its face.

- **"689 Hz, parameter-free" — RETIRED.** The once-quoted tabletop frequency was re-expressed as what it really was: a staked cutoff scale ω_c plus a parameter-free predicted *shape*. The number was never parameter-free (source: register node `rung8_falsifier`, `tier_note` and `ledger_note`).
- **The BMV entanglement-witness backup — WITHDRAWN.** An energy-basis decoherer may not degrade a position-basis witness; it has not been recomputed (source: `rung8_falsifier` `tier_note`).
- **The graviton-mass entry — demoted to vacuous.** The dispersion observable scales as m^2 , so any $m \sim H_0$ -class mass induces a phase 10^{-21} of threshold: a test that was never capable of running, not a test passed; and any such mass has Compton wavelength beyond the Hubble radius — of no observational significance in principle.

STATUS: CLOSED (demoted to vacuous — arithmetically right, evidentially empty; the $\leq 10^{-21}$ figure is a test-reach number, hard-fenced against banking as a GRUT effect) — (source: SIGNATURE_AUDIT.md item 3 and NOT-banked fence).

- **The $\mu = 4/3$ trace-only endpoint — the framework's own self-exclusion.** The modification GRUT's conformal coefficient naively suggests is excluded by the framework's own consistency analysis plus data.

STATUS: CLOSED (self-exclusion: separate-universe consistency + low- ℓ ISW) — canonical claim 14, verbatim (source: books/CORPUS_CHARTER.md ; NO_GO_LEDGER.md entry 2).

- **The registered spectral law $s = 3$ — rejected by the framework's own calculation.** The flat-scope computation returned $s = 5$ exactly, rejecting the registered value at the frozen tolerance; importing $s = 3$ anywhere is laundering, per the dispatch spec.

STATUS: DERIVED (flat scope; rejects the framework's own registered $s = 3$) — canonical claim 6, verbatim (source: books/CORPUS_CHARTER.md ; GRUT_PROGRAM_FREEZE.md §§3–4).

- **Evolving $w(z)$ as a GRUT prediction — never earned.** The sourced statement is $w = -1$ flat; any evolving $w(z)$ requires the inserted, un-sourced $\tau_2 \sim 1/H_0$.

STATUS: DERIVED (within the choices $x = 0$ / pure-TT: the sourced cosmology statement) for $w = -1$ flat; HYPOTHESIS (requires the inserted, un-sourced $\tau_2 \sim 1/H_0$, priced +2) for evolving $w(z)$ — canonical claim 13, verbatim (source: books/CORPUS_CHARTER.md ; NO_GO_LEDGER.md entry 3). Both w_a -sign over-claims — the "wrong sign vs DESI" reading and its mirror " $w_a \leq 0$ is the prediction" — were retracted; the sign is frontier-indeterminate (source: NO_GO_LEDGER.md entry 3, retraction; books/B00K_VI_COSMOLOGY.md VI.3).

- **The GW dephasing route — real but closed as unobservable.** The dissipative dephasing exists (absent in lossless GR) and sits 22–62 orders below threshold across the band; the GW170817 speed bound is satisfied with 26–66 orders to spare.

STATUS: CLOSED (computed outcome B — real-but-unobservable; FAILS-DIFFERENTIATION; Planck suppression structural, not tuned) — (source: register node `rung4_love_kk`; `calc/gw_dissipation_bounds.py`; `NO_GO_LEDGER.md` entry 5). The 2026-08-20 scope fence travels with this: the statement stands *as a dephasing statement*; the amplitude channel of a two-scale kernel is not covered by it — that channel is exactly the Γ_T slot, closed separately in §5.

- **The discriminator hunts themselves — empty, twice, plus X_FLOOR.** FOREST-EMPTY (confirmed by an independent route in phase 11); X-floor R1 returned NO PIN with the confirmation-bias trap unsprung ($x = \alpha^2$ structurally unreachable; "α-power = 0 as an output"); the dynamics route closed as a classifier (cone, no ceiling, no pin) (source: `PHYSICS_LEDGER/FOREST_PHASE11_MAPPING.md`; `calc/RESULTS_anomaly_c0_map.md`; `calc/RESULTS_x_no_pin.md`; `GRUT_PROGRAM_FREEZE.md` §3).
- **And the Γ_T candidate itself** — the record's one gated candidate, closed by computation in §5. The graveyard's newest stone is its best-documented one.

9 • Future gates and the reopening keys

The rule going forward. Any new candidate — USL shape, rung8 bookkeeping, or anything a future record proposes — requires its own eleven-step gate, designed and pre-registered before any computation, with verdicts routed per horn through the five-outcome tree. PREDICTED entry requires DISCRIMINATING on all five requirements plus survival of the program's standard screens. Absent a survivor, the standing statement remains "no discriminator identified on the current record," and the freeze's stop rule governs (source: `GRUT_PREDICTION_GATE_GAMMA_T.md` §9; `GRUT_PROGRAM_FREEZE.md` §1).

The reopening keys (any one justifies unfreezing; every vote external):

1. **O2 computed** — the interacting graviton zero-mode: *lifted* and the persistence claim falls; *protected* and the one surviving derived structure strengthens materially.
2. **The RESIDUE test decided** — the half-line/KMS alignment derived from unoriented hypotheses, or the functorial no-go proved.
3. **The SLOT test decided** — a rigorous single-patch $G_N \rightarrow 0$ limit (CLPW §4.3).
4. **Nature votes at $w = -1$** — any observed crossing excludes the entire relaxational class at a stroke.
5. **An independently motivated principle** naming a specific phenomenon the framework cannot represent — the only door for any new rung.

STATUS: UNRESOLVED (posed, decidable, unrun — held as governance conditions, not as results) — the O2 / RESIDUE / SLOT tests (source: `GRUT_PROGRAM_FREEZE.md` §5).

The nearest empirical threads, in the order the record leaves them: the DESI consolidation watch (§6 — the one place nature is currently moving toward a verdict); a dedicated QNM/ringdown calculation (converting "signature-null by inheritance" into "signature-null by calculation," or surprising everyone);

the rung8 gray-zone check (an $O(1)$ off-diagonal energy coupling would lift quiet-or-faint and make the USL shape testable); and the TT-auto instrument's owed A6 check plus any future computed x^* . None of these is a prediction; each is a place a vote could land.

What would show the whole approach was wrong — kept in this book because a test of the method is a test: a self-closing description of the observed asymmetric phenomena with zero contentful undischarged inputs would falsify the consolidated "every closure consumed an input" record outright; a demonstrated second discharge row anywhere would break the conservation pattern; and if the frozen-predicate calibration ever shows the consolidated claim deflating paradigm cases, it dies as ill-typed rather than false — the honest worst end (source: `GRUT_PROGRAM_FREEZE.md` §6).

Sectors this book cannot test because the record contains no account of them at all:

STATUS: UNMAPPED — flavor, strong-CP, neutrino masses, dark matter, baryogenesis — canonical claim 22, verbatim (source: `books/CORPUS_CHARTER.md` ; `GRUT_PROGRAM_FREEZE.md`).

The book closes where it opened. The PREDICTED section is empty, for a derived reason; the machinery that keeps it honestly empty — gates, seals, fences, and the stop rule — is the program's demonstrated contribution; and the standing posture is the framework's own closing sentence, adopted verbatim: **"Nature and mathematics hold every one of these votes. The framework holds none of them."**

Sources drawn from

- `books/CORPUS_CHARTER.md` (status vocabulary; canonical status table, claims 6, 12–16, 21–22 used verbatim)
- `GRUT_PREDICTION_GATE_GAMMA_T.md` (the eleven-step gate; decision tree; design verdict)
- `calc/SPEC_gw_tensor_friction.md` (the immutable calc-level pre-registration; four outcomes; traps; clock scoping)
- `calc/RESULTS_gw_tensor_friction.md` (the closure: REFUSE; the parameter-free pin; the conditional exhibits; the hostile verification; standing consequences)
- `SIGNATURE_AUDIT.md` (full: the table; EMPTY verdict; the 2026-08-02 items 1–5; the QNM soft spot; the NOT-banked fence and gate-to-readmit)
- `GRUT_MODEL_FRAMEWORK.md` (§§2–8; §8 for the strengthen/weaken lists and the closing posture sentence)
- `GRUT_PROGRAM_FREEZE.md` (§1 stopping rule; §3 ledger incl. PREDICTED-EMPTY reason; §4 what the audits killed; §5 reopening conditions; §6 approach-falsifiers; §7 the method)
- `GRUT_NEXT_STEPS.md` (prediction hunt; experimental bridge; stop rule)
- `NO_GO_LEDGER.md` (strength legend; entries 2–5; the no-FORBIDDEN calibration)
- `provenance/claims.json` — node `rung8_falsifier` in full (statement, tier_note, ledger_note, differentiator, cutoff_convention); register framing (74 nodes, sha256 beaeb84e8a6f8468, read-only to this corpus)
- `HOW_TO_VERIFY.md` (§3 seals; §4 what green means)
- `provenance/prereg/MANIFEST.txt` (the seal mechanism, as directory evidence)

- `STAGE_CLOSE_2026-08-09.md` and `AGENT_COORDINATION.md` (the self-certification pattern)
- `calc/RESULTS_isw_tt_auto.md` (the TT-auto gate history and amendment record)
- `calc/RESULTS_anomaly_c0_map.md`, `calc/RESULTS_x_no_pin.md`, `X_FLOOR_MAP.md`, `PHYSICS_LEDGER/FOREST_PHASE11_MAPPING.md` (the empty hunts)
- Sibling books for cross-book consistency of shared statuses:
`books/BOOK_III_QUANTUM_REALITY.md` (USL),
`books/BOOK_IV_GRAVITY_SPACETIME_RESPONSE.md` (GW observables),
`books/BOOK_VI_COSMOLOGY.md` (DESI, kernel-class discriminator)
- External literature only as the record cites it: arXiv:2507.03103 (SCDP open EFT / the slot bound); arXiv:2503.14738 (DESI DR2+CMB); arXiv:1305.5231 (Anastopoulos–Hu, via Book III's citation of the record)

Gaps in this book

1. **No gate design exists for the USL shape candidate** — the record names it as a candidate and this book cannot supply its eleven-step design; doing so would be new work, not corpus assembly.
2. **No gate design exists for the rung8 bookkeeping candidate** — same absence.
3. **The record does not expand the acronym USL** — carried here, as in Book III, as the register's name for the rung8 shape signature.
4. **No dedicated QNM/ringdown calculation** — the signature audit's one explicit caveat; "signature-null by inheritance" remains an argument, not a computation.
5. **The $\leq 10^{-21}$ tabletop/GW figures remain un-readmitted** — no calculation exists behind them; the gate that could have readmitted them closed as a computed refusal.
6. **EDIT 1's conditional marker is unfinalised** — a REFUSE is not the scalar-only Q-A answer that finalisation required.
7. **The TT-auto gate's A6 Boltzmann-grade differential check is owed** before any $\kappa=3$ kill-grade use, and no computed candidate x^* exists for the gate to adjudicate.
8. **The amplitude channel of the two-scale kernel is uncovered** and the ω_c adjudication is owed — three in-corpus values span 39.6 orders; the crossover moves ~ 10 orders on that choice alone (shared with Book IV's gap list). The ~ 10 -order and 19.8-order figures are consistent: crossover $\propto \sqrt{\omega_c}$ (SPEC §4), so the full span halves to 19.8 orders while adjacent pairs move ~ 10 .
9. **The DESI consolidation is external** — the record can fence its attribution but cannot decide it; this book reports the audit's standing disposition only.
10. **R5's "named future sensitivity" clause has no named instrument on the record** — the one candidate forecast ("ET closes ~ 1.5 orders") is explicitly NOT banked; any future gate needing a sensitivity name must source one first.
11. **The rung8 gray-zone check is pending** — the one operator-algebra question whose answer could lift quiet-or-faint is posed in the register and unanswered.
12. **Whole sectors are UNMAPPED** (flavor, strong-CP, neutrino masses, dark matter, baryogenesis) — no GRUT account exists, hence no test content; recorded here so their absence from this book is read as the record's absence, not an oversight.